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Studierichting: Informatica
Oefeningenreeks 1E nr. 6

$$A(z) = (z-2)(z-3)(z-4)Q(z) + az^2 + bz + c$$

$$\begin{cases} A(2) = 4a + 2b + c = 8 \\ A(3) = 9a + 3b + c = 18 \\ A(4) = 16a + 4b + c = 30 \end{cases}$$

$$\Rightarrow \begin{cases} 4a + 2b + (30 - 16a - 4b) = 8 \\ 9a + 3b + (30 - 16a - 4b) = 18 \\ c = 30 - 16a - 4b \end{cases}$$

$$\Rightarrow \begin{cases} -12a - 2b = -22 \\ -7a - b = -12 \\ c = 30 - 16a - 4b \end{cases}$$

$$\Rightarrow \begin{cases} a = -\frac{1}{6}b + \frac{11}{6} \\ -7a - b = -12 \\ c = 30 - 16a - 4b \end{cases}$$

$$\Rightarrow \begin{cases} -7\left(-\frac{1}{6}b + \frac{11}{6}\right) - b = -12 \\ a = -\frac{1}{6}b + \frac{11}{6} \\ c = 30 - 16a - 4b \end{cases}$$

$$\Rightarrow \begin{cases} \frac{1}{6}b - \frac{77}{6} = -12 \\ a = -\frac{1}{6}b + \frac{11}{6} \\ c = 30 - 16a - 4b \end{cases}$$

$$\Rightarrow \begin{cases} b = 5 \\ a = 1 \\ c = -6 \end{cases}$$

$$\Rightarrow R(z) = z^2 + 5z - 6$$

References